



Distributed systems

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Consensus



Fault-tolerant total order broadcast

- Total order broadcast is very important for State Machine Replication
  - It can be implemented by marking one node as a ***leader*** and sending all messages through the leader
  - The leader then uses FIFO total-order broadcast to sync with replicas
  - Leader is SPOF
  - Can we transfer ***leadership*** to another node automatically? → consensus algorithms

Consensus problem

- Several nodes want to come to an ***agreement*** about a value
- One or more nodes can propose, and the consensus algorithm will decide
- Consensus algorithm will ensure that a value is chosen from proposed values, and that all nodes agree on the same value

Consensus vs T.O.B

- What is the relationship between **consensus** and **total-order broadcast?**
  - Interchangeable concepts, formally equivalent [Chandra & Toueg, 1996]
  - T.O. Broadcast → consensus: messages are proposed values, and the first message delivered is accepted
  - Consensus → T. O. Broadcast: we use a consensus protocol to agree on the first, second, third, … message

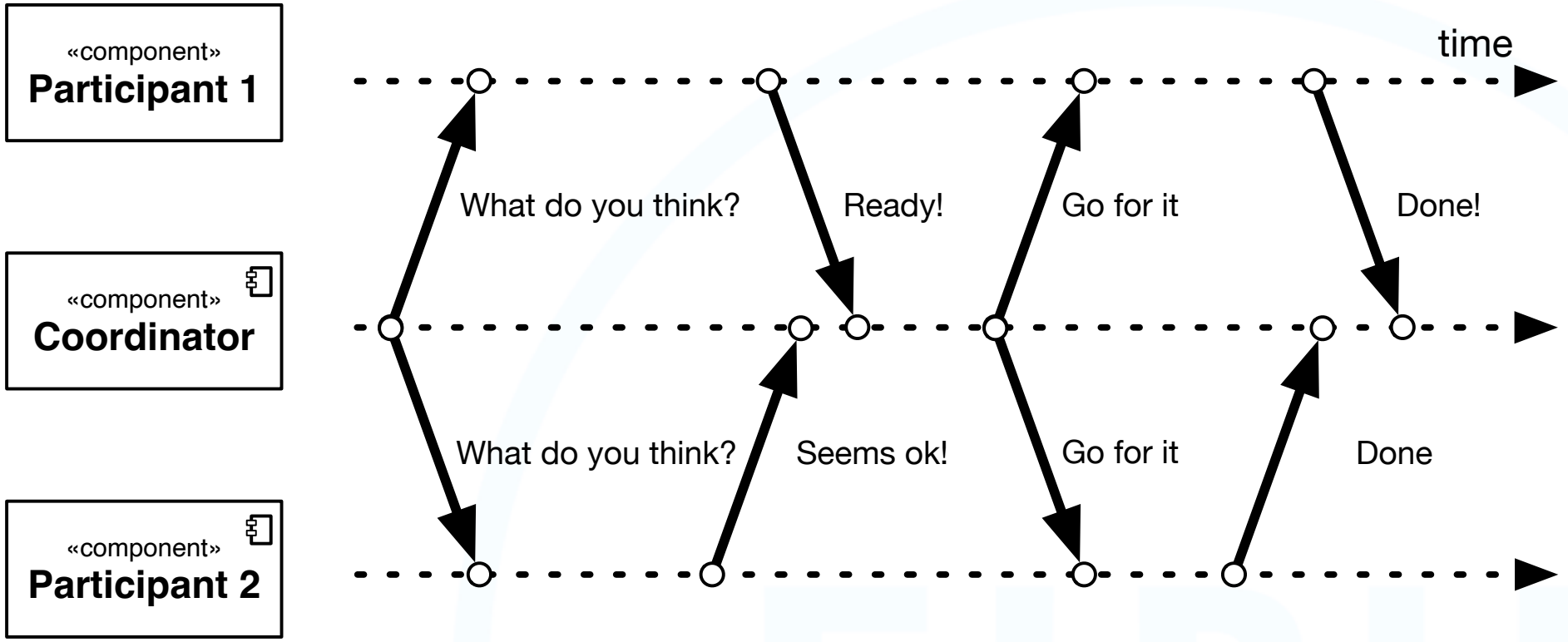
Popular consensus algorithms

- **Paxos:** single-value consensus
- **Multi-Paxos:** generalization to total-order broadcast
- **Raft** (Ongaro&Ousterhout, Stanford)**, Viewstamped Replication** (B. Liskov&Cowling, MIT)**, Zab** (FIFO total-order by default)

Primitive: 2PC

- **Two-phase commit**
  - A necessary primitive to achieve any consensus algorithm
  - Used widely in distributed, fault-tolerant databases for actual transaction **commits**
  - **Here will be described in general sense**

2 Phase Commit



Self-assesment

Why is consensus important?